



```
moisespr123@moisespr123-Lenovo-IdeaPad-Y510P: ~  
File Edit View Search Terminal Help  
moisespr123@moisespr123-Lenovo-IdeaPad-Y510P: $ cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor  
performance  
performance  
performance  
performance  
performance  
performance  
performance  
performance  
moisespr123@moisespr123-Lenovo-IdeaPad-Y510P: $
```

CHANGING THE CPU GOVERNOR TO PERFORMANCE IN LINUX

📅 September 18, 2022 👤 Moises Cardona 💬 1 Comment

Hi everyone,

In this post, I'll show you how to change the CPU governor in Linux to the Performance mode. This will allow your machine to run to its full potential as the CPU will be running to its maximum supported clock regardless of the load.

Normally, we can choose between these CPU scaling governors:

- performance: Sets the CPU to the highest supported clock speed.
- powersave: Sets the CPU to the lowest supported clock speed.
- userspace: Allows the root user to set the cpu speed manually by creating a configuration file called `scaling_setspeed`.
- ondemand: Dynamically set the CPU speed based on the CPU load.

Therefore, if we want to get the maximum performance, we should use the `performance` governor.

Some Linux distributions may also use another governor called `schedutil` which should communicate better to the hardware to set the CPU frequency dynamically, but you may want to switch to `performance` if you want to bypass this. As a downside of doing this is

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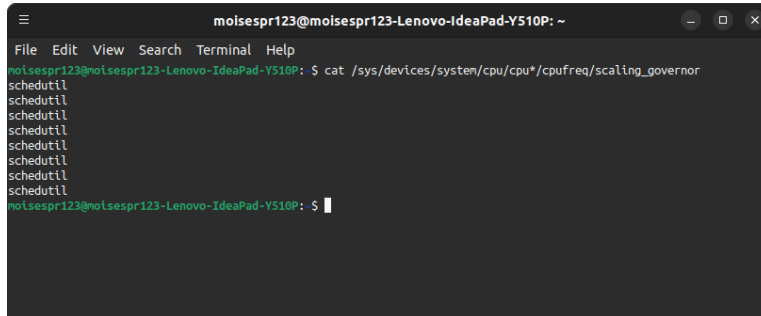
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You can verify the CPU governor currently in use by checking the content of the `scaling_governor` file. This is located in `/sys/devices/system/cpu/cpu{cpu core or thread number}/cpufreq/`

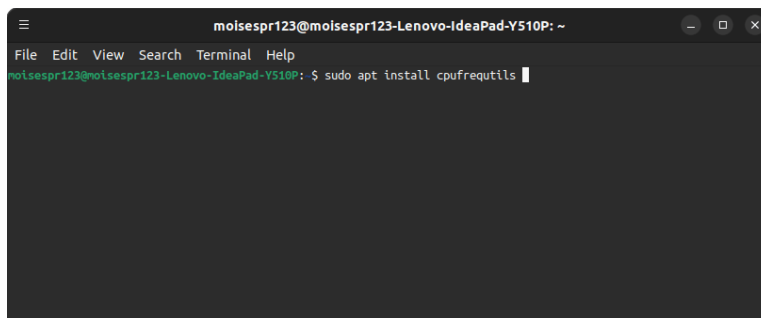
To print the governor of every CPU thread, you can use an asterisk wildcard (*) when passing the CPU core or thread number when using the `cat` command:

```
cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor
```



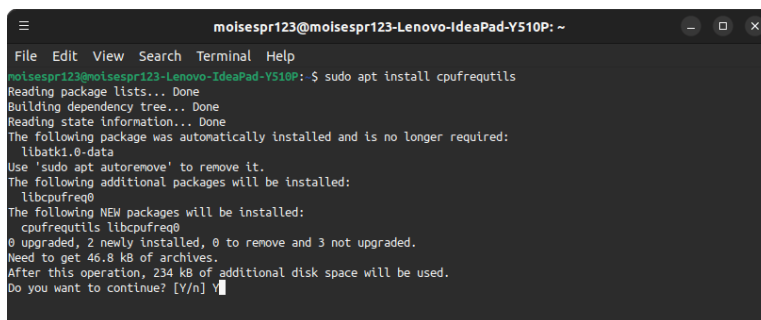
```
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File Edit View Search Terminal Help  
moisespr123@moisespr123-Lenovo-IdeaPad-Y510P: $ cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor  
schedutil  
schedutil  
schedutil  
schedutil  
schedutil  
schedutil  
schedutil  
schedutil  
moisespr123@moisespr123-Lenovo-IdeaPad-Y510P: $
```

To change our CPU governor to `performance`, we must have the `cpufrequtils` package installed in our system. To install it, type `sudo apt install cpufrequtils`:



```
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moisespr123@moisespr123-Lenovo-IdeaPad-Y510P: $ sudo apt install cpufrequtils
```

Confirm the installation with the Y key and pressing Enter:



```
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File Edit View Search Terminal Help  
moisespr123@moisespr123-Lenovo-IdeaPad-Y510P: $ sudo apt install cpufrequtils  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following package was automatically installed and is no longer required:  
  libatk1.0-data  
Use 'sudo apt autoremove' to remove it.  
The following additional packages will be installed:  
  libcputime0  
The following NEW packages will be installed:  
  cpufrequtils libcputime0  
0 upgraded, 2 newly installed, 0 to remove and 3 not upgraded.  
Need to get 46.8 kB of archives.  
After this operation, 234 kB of additional disk space will be used.  
Do you want to continue? [Y/n] Y
```

The installation should now complete:

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```
moisespr123@moisespr123-Lenovo-IdeaPad-Y510P: ~$ sudo apt install cpufrequtils
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libatk1.0-data
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libcpufreq0
The following NEW packages will be installed:
  cpufrequtils libcpufreq0
0 upgraded, 2 newly installed, 0 to remove and 3 not upgraded.
Need to get 46.8 kB of archives.
After this operation, 234 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://us.archive.ubuntu.com/ubuntu kinetic/universe amd64 libcpufreq0 amd64 008-2build1 [12.3 kB]
Get:2 http://us.archive.ubuntu.com/ubuntu kinetic/universe amd64 cpufrequtils amd64 008-2build1 [34.5 kB]
Fetched 46.8 kB in 0s (125 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libcpufreq0.
(Reading database ... 525554 files and directories currently installed.)
Preparing to unpack .../libcpufreq0_008-2build1_amd64.deb ...
Unpacking libcpufreq0 (008-2build1) ...
Selecting previously unselected package cpufrequtils.
Preparing to unpack .../cpufrequtils_008-2build1_amd64.deb ...
Unpacking cpufrequtils (008-2build1) ...
Setting up libcpufreq0 (008-2build1) ...#####
Setting up cpufrequtils (008-2build1) ...
update-rc.d: warning: start and stop actions are no longer supported; falling back to defaults
update-rc.d: warning: start and stop actions are no longer supported; falling back to defaults
Processing triggers for man-db (2.10.2-2) ...
Processing triggers for libc-bin (2.36-0ubuntu2) ...
moisespr123@moisespr123-Lenovo-IdeaPad-Y510P: ~$
```

Next, we will write `GOVERNOR="performance"` to the `cpufrequtils` configuration file which will make our CPU be always set to `performance`. There are multiple ways to write this, but we can easily type the following in the terminal window to write it:

```
echo 'GOVERNOR="performance"' | sudo tee /etc/default/cpufrequtils
```

Note the single and double quotes.

Press Enter and it will print the echo line to the screen, while writing it to the `/etc/default/cpufrequtils` file.

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moisespr123@moisespr123-Lenovo-IdeaPad-Y510P: ~$ echo 'GOVERNOR="performance"' | sudo tee /etc/default/cpufrequtils
GOVERNOR="performance"
moisespr123@moisespr123-Lenovo-IdeaPad-Y510P: ~$
```

If we were to check our CPU governor now, we will see that it is set to `ondemand` as a consequence of installing `cpufrequtils`.

```
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File Edit View Search Terminal Help
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GOVERNOR="performance"
moisespr123@moisespr123-Lenovo-IdeaPad-Y510P: ~$ cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor
ondemand
ondemand
ondemand
ondemand
ondemand
ondemand
ondemand
ondemand
ondemand
moisespr123@moisespr123-Lenovo-IdeaPad-Y510P: ~$
```

To change it to our newly defined `performance` governor, we will restart the `cpufrequtils` service:

```
sudo systemctl restart cpufrequtils.service
```

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GOVERNOR="performance"
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moisespr123@moisespr123-Lenovo-IdeaPad-Y510P: ~$ sudo systemctl restart cpufrequtils.service
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GOVERNOR="performance"  
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ondemand  
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moisespr123@moisespr123-Lenovo-IdeaPad-Y510P: $ sudo systemctl restart cpufrequtils.service  
  
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And with this, we have concluded changing our Ubuntu default CPU governor.

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